



Sadir Ahmed Zidan

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AI/ML Engineer with 2+ years of production experience building and deploying LLM-powered applications. Specialized in fine-tuning LLMs with LoRA/QLoRA, designing advanced RAG & agentic systems, and delivering low-latency inference APIs using LangChain, Groq, FastAPI, Docker, and MLflow. Hands-on expertise in computer vision, 3D reconstruction and multilingual retrieval pipelines (FAISS, Milvus), along with voice enabled systems. Proven track record of shipping scalable AI solutions end-to-end, from custom model training and pretrained model integration to production deployment and monitoring.

Education

American International University Bangladesh

BSc in Computer Science & Engineering, January 2020 - August 2024

CGPA: 3.77/4.00

Work Experience

Machine Learning Engineer - Fulltime, Onsite

Nasir Syntax Solution Limited (concern of Nasir Group), January 2024 - Present

- Designed and shipped production RAG and agentic LLM systems using LangChain, FAISS, Milvus, and custom retrieval pipelines for vector search, personalized search and custom chatbots. reduced retrieval latency by 40% through pipeline optimization and index tuning.
- Fine-tuned and quantized LLMs with LoRA/QLoRA, built dynamic model loading and inference optimization workflows, achieved 60% reduction in model memory footprint enabling deployment on resource-constrained and edge-compatible environments.
- Developed computer vision pipelines including object detection, segmentation, ocr for document data extraction and 3D reconstruction projects.
- Trained custom deep learning models from scratch and integrated pretrained models.
- Built and deployed scalable FastAPI inference services with Docker, MLflow tracking, request validation, rate-limiting, and monitoring.

AI Developer - Part time, Remote

GradMate.ai, Apr 2025 - Feb 2026

- Built high-performance agentic RAG systems and multi-step AI workflows using LangChain and Groq's ultra-fast LLM inference.
- Implemented real-time speech-to-text (STT) and text-to-speech (TTS) integrations into educational AI agents.
- Optimized latency and cost for Groq/LangChain pipelines serving live user-facing features, cut average inference response time by 35% and reduced per-query API cost through prompt compression and model routing strategies.
- Collaborated on modular AI architecture, data contracts, and retrieval quality standards with frontend and platform teams.

AI Engineer - Contractual, Remote

Ambitionix, May 2025 - Jan 2026

- Fine-tuned LLMs for specific tasks.
- Created agents and APIs for deployment.
- Collaborated with cross-functional teams to achieve project goals.

Machine Learning Intern

Nasir Syntax Solution Limited (concern of Nasir Group), October 2023 - December 2023

- Conducted research on personalized recommendation systems, chatbots, and text generation using LLMs and computer vision.
- Scraped and preprocessed data, including tokenization and vectorization, and fine-tuned LLMs for specific applications.

Research

- **Bachelor's Thesis - A Study on the Use of Machine Learning Algorithms for Predicting Mental Health Conditions**

American International University Bangladesh, 2023 | Not Published

Investigated classical ML approaches (SVM, Random Forest, ensemble methods) on structured clinical/survey data for early mental health condition prediction. Explored feature selection, class imbalance handling, and model interpretability.

- **Research Paper (Manuscript in Preparation) - Efficient Monocular Depth Estimation for Edge Devices (working title)**

Co-author | Nasir Syntax Solution Limited | Ongoing, 2025 - Present

Contributing to experiments and manuscript writing for a study on lightweight monocular depth estimation models optimized for edge deployment. Focus on minimizing computational resource usage while maintaining competitive accuracy against state-of-the-art methods.

Target venue: international conference / journal (TBD).

Skills

- **Programming Languages:** Python, R
- **Frameworks & Libraries:** TensorFlow, Keras, LangChain, Groq, PyTorch, Fast API, Pandas, Faiss, Llama cpp, ONNX, Ollama, Hugging Face, Streamlit
- **Tools & Platforms:** Docker, MLFlow, Kubernetes, GitHub, VS Code, R Studio, Colab, LoRA
- **Databases:** SQL, MongoDB
- **Operating Systems:** Windows, Linux

Personal Projects

- **Multilingual RAG System for Bengali and English Queries**
 - A system to process queries in English and Bengali, retrieve relevant information from a PDF document (specifically the HSC26 Bangla 1st Paper book), and generate meaningful answers.

- **Face Mask Detection**
 - A system to detect facial masks using a cascade classifier. Used a large dataset that varies with different types of poses with masks worn by humans. The model got good accuracy after implementation.
- **Drowsiness Detection**
 - A real-time system using machine learning algorithms and image processing to detect driver drowsiness. It can continuously monitor the driver while they are driving and provide immediate feedback or alerts.
- **Sentiment Analysis Chatbot**
 - This project demonstrates a simple sentiment analysis chatbot that analyzes the user's mood based on their input and provides contextual responses.
- **Life Insurance Support Assistant**
 - Agentic life-insurance assistant built using LangChain + Groq, designed to answer domain-specific questions about policy types, benefits, eligibility, and claims, using structured category-based reasoning.
- **Voice-Enabled Agentic RAG**
 - A modular, local Retrieval-Augmented Generation (RAG) pipeline for multi-domain document Q&A, featuring PDF cleaning, adaptive chunking, FAISS vector indexing, multi-agent retrieval, and both text and voice CLI interfaces.